



Republic of Rwanda
Ministry of Education



RTB | RWANDA
TVET BOARD

SOLIDWORKS DESIGNS

GENSD401

Produce SolidWorks Design

Competence

RQF Level: 4

Learning Hours



60

Credits: 6

Sector: ICT and Multimedia

Trade: Computer System and Architecture

Module Type: General

Curriculum: ICTCSA4001 TVET Certificate IV in Computer System and
Architecture

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1200

Issue Date: September, 2023

Purpose statement	This module describes employable and practicable skills, knowledge and right attitudes required by the learner to Set up Solid work space, Create Part, Assemble Parts, Produce drawing for any electronic enclosure in SolidWorks software.						
Learning assumed to be in place	- Fundamental Mathematics Analysis - Digital electronics						
Delivery modality	Training delivery			100%	Assessment		Total 100%
	Theoretical content			30%	Formative assessment	30%	50%
	Practical work:			70%		70%	
	Group project and presentation	20%					
	Individual project /Work	50%					
				Summative Assessment		50%	

Elements of Competency and Performance Criteria

Element of competence	Performance criteria
1. Set up Solid work space	1.1. Solid Works software is properly installed according to the manufacturer's instructions
	1.2. Template is correctly selected according to the work required
	1.3. Interface is properly customized according to the work to be done
2. Create Part	2.1. Plane is properly selected according to part orientation
	2.2. Sketch is correctly created according to the required geometry
	2.3. Apply features are properly applied according to the required shape
	2.4. Visualization is properly applied visualization according to the required part state

	2.5. Part is correctly exported according to the required format
3. Assemble Parts	3.1. Parts are imported according to the assembly layout
	3.2. Mating of parts and components and is properly applied according to the assembly layout
	3.3. Assembly is correctly exported according to the required format
4. Produce drawing	4.1 Drawing sheet layout is correctly set according to the work to be done
	4.2 Title block is properly edited according to the required information
	4.3 Views are properly imported according to the work to be done
	4.4. Visualization is correctly applied according to the required information
	4.5. Drawing is correctly exported according to the required format

Intended Knowledge, Skills and Attitude

Knowledge	Skills	Attitude
✓ Describe measurement techniques concepts ✓ Identify Solid modeling ✓ Describe Solid work design principles ✓ Identify 3D Modeling ✓ Identify engineering drawing Principles ✓ Describe solidWorks features	✓ Create solid work diagram ✓ Import components of assemble parts ✓ Export assembly of assemble parts ✓ Apply mates ✓ Set drawing sheet layout ✓ customize Interfaces ✓ Install SolidWorks software ✓ Apply 3D Modeling design	✓ Passion ✓ Critical thinking ✓ Decision Making ✓ Problem Solving ✓ Teamwork ✓ Creativity ✓ Adaptability

Course content

Learning outcomes

At the end of the module the learner will be able to:

- 1.Set up Solid work space
- 2.Create Part
- 3.Assemble Parts
- 4.Produce drawing

Learning outcome 1: Set up Solid work space

Learning hours: 10

Indicative content

- **Installation of SolidWorks software**

- ✓ Introduction Solid Works software



Definition of key terms (CAD, CAM, CADD, CAE, 2D modelling, 3D modelling)



Application of SolidWorks



Advantages and limitations of SolidWorks

- ✓ Steps for installing SolidWorks software



Computer System Requirements Check



Opening SolidWorks setup (.EXE file)



Following installation Wizards' windows



Restarting the computer

- **Selection of Solid works template**

- ✓ Part template
- ✓ Assembly template
- ✓ Drawing template

- **Interface customization**

- ✓ Toolbars and Menus

- ✓ Keyboard Shortcuts
- ✓ Mouse Gestures
- ✓ Command Manager Customization
- ✓ Shortcut Bars
- ✓ Custom Colours and Themes
- ✓ Workspace Customization
- ✓ File Locations
- ✓ Custom Property Tab Builder

Resources required for the learning outcome

Equipment	▪ Computer ▪ Projector ▪ Whiteboard
Materials	▪ Papers ▪ Internet connectivity ▪ Whiteboard markers
Tools	▪ SolidWorks software ▪ CD Drive ▪ Flash Disk Drive ▪ SolidWorks tutorials
Facilitation techniques	▪ Trainer guide ▪ Brain storming
Formative assessment methods	▪ Performance assessment ▪ Written assessment

Indicative content

- **Selecting of plane**

- ✓ Standard plane



Front plane



Top plane



Right plane

- ✓ Custom plane

- **Creating sketch**

- ✓ 2D sketch

- ✓ Sketch tools

- ✓ Sketch entities

- ✓ Sketch relations

- ✓ Status of a Sketch



Full defined



Under defined



Over defined

- **Applying part features**

- ✓ Feature tools



Extruded boss



Extruded cut



Revolved boss



Revolved cut



Swept boss



Lofted boss



Fillet



Chamfer



Rib



Shell



Linear pattern



Circular pattern



Mirror

- **Applying Visualization**

- ✓ Material
- ✓ Appearance
- ✓ Scene and environment
- ✓ View orientation
- ✓ Display style
- ✓ Lighting
- ✓ Camera setting
- ✓ Evaluate

- **Exporting part**

- ✓ File location
- ✓ File format

Resources required for the indicative content

Equipment

- Computer
- Projector
- Whiteboard

Materials

- Student workbook
- Papers

	▪ Internet connectivity
Tools	▪ SolidWorks Software ▪ Tutorials ▪ Online Courses
Facilitation techniques	▪ Trainer guided ▪ Group discussion
Formative assessment methods	▪ Product assessment ▪ Written assessment

Learning outcome 3: Assemble parts		Learning hours:10
Indicative content		
• Importing components/inserting components. ✓ Make assembly from a part ✓ Select "Component" and then choose "Existing Part/Assembly." ✓ Browsing components • Applying mates ✓ Standard mates ✓ Advanced mates ✓ Mechanical mates ✓ Analysis • Exporting assembly ✓ File location ✓ File format		
Resources required for the indicative content		
Equipment	▪ Computer ▪ Printer ▪ Projector ▪ Whiteboard	
Materials	▪ Internet bundles ▪ Papers	
Tools	▪ SolidWorks Software	
Facilitation techniques	▪ Group discussion ▪ Trainer guided	
Formative assessment methods	▪ Written assessment ▪ Performance assessment	

Indicative content

- **Setting drawing sheet layout**

- ✓ Sheet size



Standard



Custom

- ✓ Orientation



Land scape



Portrait

- **Editing the title block**

- ✓ Editing table

- ✓ Editing information



Elements of title block

- **Importing views**

- ✓ Updating view palette

- ✓ Model view

- **Applying Visualization**

- ✓ drawing views



Types of drawing views

- ✓ Annotation

- ✓ Display

- ✓ Scale

- **Exporting the drawing**

- ✓ File location

- ✓ File format

Resources required for the indicative content

Equipment	▪ Computer ▪ Printer ▪ Projector
Materials	▪ Internet connectivity ▪ Papers
Tools	▪ Solid works software ▪ Tutorials ▪ Books
Facilitation techniques	▪ Trainer guided ▪ Group discussion
Formative assessment methods	▪ Product assessment

References:

1. <https://www.javelin-tech.com/blog/2021/02/how-to-install-solidworks-software/>
2. <https://www.goengineer.com/blog/apply-new-solidworks-drawing-sheet-format-future-drawings>
3. <https://www.takeoffpros.com/2020/03/25/using-2d-or-3d-for-your-site-models/>
4. SolidWorks Essentials (concord, Massachusetts 01742 USA)
5. <https://www.google.com/search?q=%E2%80%A2+Inserting+components+in+solidWorks&oq=%E2%80%A2%09Inserting+components+in+solidWorks&aqs=chrome..69i57j0i22i30l2.14053j0j7&sourceid=chrome&ie=UTF-8>